

AgonProject: An Online Platform for Exploring Approaches to Formal Argumentation

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TweetyProject is

- a **collection of** Java **libraries** implementing approaches to knowledge representation and reasoning:
- classical and non-classical logics,
- computational argumentation,
- logic programming,
- agents,
- belief dynamics,
- ...

agon /ăg'ŏn, -ŏn, ä-gŏn'/'



noun

1. A conflict, especially between the protagonist and antagonist in a work of literature.
2. The part of an ancient Greek drama, especially a comedy, in which two characters engage in verbal dispute.
3. A test of will; a conflict.

The American Heritage® Dictionary of the English Language, 5th Edition • More at [Wordnik](#)

Framework Construction and Visualisation

Interactive graph editor with automatic layouts and physics simulation

Semantical Reasoning

Extension- and ranking-based semantics, with enumeration and credulous / sceptical acceptance.

Export and Sharing

Export to $\text{\LaTeX}$, SVG, TGF, and the ICCMA AF format, or share by URL

Random Generation

Random frameworks from 8 configurable graph generators

Tutorials and References

10+ tutorials and a glossary with definitions and literature references

The zoo of formal argumentation, in one browser tab

AgonProject

The platform to explore different approaches to formal argumentation

[AIG Hagen](#) [TweetyProject](#) [Source cec42fd](#) [Third-Party](#)

Abstract Argumentation

The foundational model for the formal representation of argumentation via arguments and directed attacks.

OPEN EXAMPLE

meal_wine	two_cycles
grounded_ideal	initial_sets

NEW

[Create new](#) [Generate AF](#)

Bipolar Argumentation

Relations between arguments can be either attacking or supporting.

OPEN EXAMPLE

meal_wine	vacation_planning
murder_trial	

NEW

[Create new](#) [Generate BAF](#)

Dialectical Argumentation

Relations between arguments are modelled via propositional acceptance conditions.

OPEN EXAMPLE

meal_wine	murder_trial
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NEW

[Create new](#) [Generate ADF](#)

Incomplete Argumentation

Distinguish between certain and uncertain arguments and attacks to represent incomplete information.

OPEN EXAMPLE

murder_trial

NEW

[Create new](#) [Generate iAF](#)

Probabilistic Argumentation

Arguments and attacks can be assigned a probability value between 0 and 1 to quantify uncertainty.

OPEN EXAMPLE

murder_trial

NEW

[Create new](#) [Generate PAF](#)

Argumentation with Collective Attacks

Extends abstract argumentation by allowing sets of arguments to collectively attack a target argument.

OPEN EXAMPLE

tandem_trip

NEW

[Create new](#) [Generate SetAF](#)

Extension & Ranking Semantics

Stable · Enumerate

SEMANTICS
Stable

MODE
Enumerate

Stable Semantics
A set of arguments E is a stable extension iff E is conflict-free and we have that $E \cup E^+ = A$.

{Fish, Red} **{Fish, White}** **{Meat, Red}**

{Pasta, Red}

1ms · Select extension to highlight

Copy results

Categorizer · Ranking

SEMANTICS
Categorizer

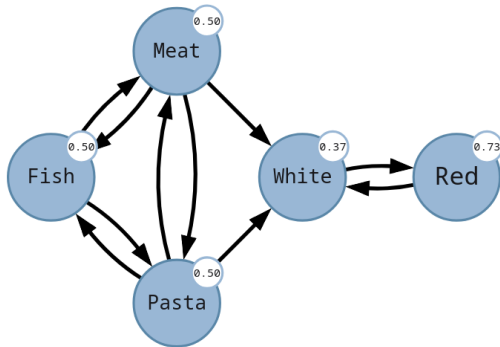
Epsilon
0.001

Categorizer Ranking
The categorizer ranking associates to any AF F a ranking $\succeq_F^{\text{Cat}}$ on A such that $\forall a, b \in A : a \succeq_F^{\text{Cat}} b$ iff $\text{Cat}_F(a) \geq \text{Cat}_F(b)$, where Cat_F is the categorizer function.

Pasta 0.5 **Fish 0.5** **Meat 0.5** **White 0.37** **Red 0.73**

1ms · Node labels show ranking scores

Copy ranking



Classical	conflict-free, admissible, complete, grounded, preferred, stable
Admissibility-based	ideal, semi-stable, eager, strong admissibility, initial sets, ...
Non-admissible	naive, CF2, SCF2, stage, stage2, (strongly) undisputed
Meta-families	weak admissibility, (semi-)qualified, vacuous-reduct, serialisable
Argument-rankings	burden, categoriser, counting, discussion, iterated graded defense, probabilistic, propagation, serialisability, strategy-based, ...

Stable · Enumerate

SEMANTICS

Stable

MODE

Enumerate

Stable Semantics (ADF)

A three-valued interpretation v of an ADF is a stable model iff $v^{-1}(t) = v_{gr}^{-1}(t)$, where v_{gr} is the grounded model of the reduced ADF $D_{|v}$.

{Guilty, ¬Innocent, Motive, ¬Alibi, Evidence}

{¬Guilty, Innocent, Motive, Alibi, ¬Evidence}

4ms · Select model to highlight

Copy results

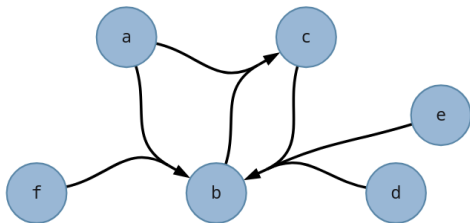
The diagram illustrates an argumentation framework with five nodes: Motive (green), Guilty (green), Innocent (red), Evidence (green), and Alibi (red). The edges and their associated formulas are as follows:

- Motive → Innocent: $\text{Alibi} \wedge \neg \text{Evidence} \vee \neg \text{Motive}$
- Innocent → Guilty: $\neg \text{Innocent}$
- Evidence → Innocent: $\neg \text{Alibi}$
- Alibi → Innocent: $\neg \text{Evidence}$
- Evidence ↔ Alibi: $\neg \text{Alibi}$ (from Evidence to Alibi) and $\neg \text{Evidence}$ (from Alibi to Evidence)

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Collective attacks & Exporting



Export

Format LaTeX (argumentation) 

Style Parameters 

`\usepackage[namestyle=math]{argumentation}` 

 Save text  Copy text  Save SVG  Copy SVG

```
1 \begin{af}
2   \argument{a1}{a} at (1.00,1.73)
3   \argument{a2}{b} at (2.00,0.00)
4   \argument{a3}{c} at (3.00,1.73)
5   \argument{a4}{d} at (4.00,0.00)
6   \argument{a5}{e} at (4.50,0.87)
7   \argument{a6}{f} at (0.00,0.00)
8   \setattack{a1,a6}{a2}
9   \setattack{a1,a2}{a3}
10  \setattack{a3,a4,a5}{a2}
11 \end{af}
```

Dedicated mobile UI

vacation_plan (BAF)

Deductive Support
Under the deductive support interpretation, an argument a supports an argument b if the acceptance of a logically implies the acceptance of b .

Preferred Semantics
A set of arguments E is a preferred extension iff E is complete and $\subseteq$ -maximal.

{Beach, Flight}
{City, Flight}
{Mountains, Roadtrip, Train}

AgonProject

Generate random BAF

Algorithm
Erdos Renyi

N
10

P
0.3

TYPE OPTIONS

Support Perc
0.3

Self Loop Prob
0

Fixed seed

LAST GENERATED
10 arguments, 21 attacks, 7 supports.

Open in editor **Download TGF**

Generate

Glossary

AF BAF **ADF** iAF PAF SetAF

Search terms...

A

Abstract Dialectical Framework
An abstract dialectical framework $D = (A, L, C)$ consists of a set of arguments A , a set of links $L \subseteq A \times A$, and for each argument $a \in A$ an acceptance condition C_a over its parents $\text{par}(a)$.

Acceptance Condition
An acceptance condition C_a for argument a in an ADF is modelled as a propositional formula over the parents of a that essentially determines when a is accepted given the set of accepted parents.

Admissibility (ADF)
A three-valued interpretation v of an ADF is an admissible model iff $v \leq_i \Gamma_D(v)$, where $\Gamma_D(v)$ is the characteristic operator for ADFs.

C

Characteristic Operator
The characteristic operator Γ_D of an ADF D is defined as $\Gamma_D(v)(a) = \bigcap \{v'(C_a) \mid v' \in [v]_2\}$ for all $a \in A$. It maps v to the consensus over its two-valued completions.

murder_trial (IAF)

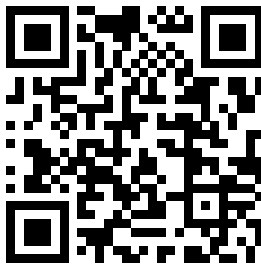
Step 3 of 6

Create an uncertain argument
Switch to the **dashed circle** mode, then **double-tap** on the canvas to create an uncertain argument. It appears with a dashed border.

< Back **• Waiting for you...**

Evaluate

- more formalisms, e.g., **ABA**, **weighted AFs**, **BSAFs**, **claim-augmented AFs**
- benchmark generators from the ICCMA ecosystem
- other reasoning approaches, e.g., acceptance explanations



`agon.tweetyproject.org`